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With sincere respect,

Anish Nepal

Bigyan Khatri

Unique Oli

ABSTRACT

HamroPasal is a localized web application designed to modernize inventory management for Nepal's small retail (kirana) shopkeepers, who form the backbone of the country's retail sector. Built on the Laravel MVC framework, the system prioritizes flexibility, reusability, and extensibility to deliver a user-friendly platform tailored to low-tech environments. Key features include real-time inventory tracking, sales recording, and automated stock-level updates to prevent overstocking or shortages. The application offers role-based access (Shopkeeper/Admin), a unified dashboard for inventory insights, and secure authentication—all optimized for Nepali currency and literacy levels. This project leverages Laravel (PHP) and MySQL to ensure scalability, data security (HTTPS, encryption), and seamless performance. By replacing error-prone manual logs and spreadsheets, **HamroPasal** streamlines operations, reduces human errors, and helps shopkeepers maintain optimal inventory levels. The documentation covers system architecture, feature modules, and competitive advantages over traditional methods, positioning **HamroPasal** as an essential tool for Nepal's micro-retailers

Keywords: Inventory Management, Laravel MVC, Small Business Digitization, Real-time Tracking, User-centric Design

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CHAPTER 1

INTRODUCTION

1.1. Project Introduction

HamroPasal is an inventory management web application designed to transform how Nepal's small retail (kirana) shopkeepers track stock, manage suppliers, and streamline daily operations. The system replaces outdated manual logs and error-prone spreadsheets with a user-friendly digital platform, offering real-time inventory updates, sales recording, and actionable insights. With **HamroPasal**, shopkeepers can minimize stock outs, reduce excess inventory, and make data-driven decisions to enhance profitability. The application features Product and Category Management, enabling users to add, edit, or remove items effortlessly, while Supplier Management ensures seamless tracking of purchase history. A centralized Dashboard provides an overview of stock levels, sales trends, and critical alerts, empowering shopkeepers to act swiftly. The system also includes Role-Based Access, with dedicated interfaces for Shopkeepers (Users) and Administrators, ensuring secure and efficient operations tailored to each user's needs. Built on the Laravel MVC framework with MySQL, **HamroPasal** prioritizes scalability, security (HTTPS, encrypted data), and localization—supporting Nepali currency and low-literacy users. By automating inventory workflows and eliminating manual errors, the system saves time, reduces waste, and helps small businesses thrive in Nepal's competitive retail market. **HamroPasal** is more than a tool; it's a step toward digitizing Nepal's micro-retail sector, empowering shopkeepers to focus on growth rather than paperwork. With its intuitive design and robust features, the system bridges the gap between traditional practices and modern efficiency, creating a sustainable solution for the backbone of Nepal's economy.

1.2. Problem Statement

Small retail shops in Nepal rely heavily on manual inventory tracking methods, such as paper logs or basic spreadsheets, which are inefficient and error-prone. These outdated systems lead to inaccurate stock records, frequent overstocking or stock outs, and wasted time on repetitive data entry. Without real-time visibility into inventory levels, shopkeepers

struggle to make informed purchasing decisions, resulting in lost sales and reduced profitability.

The lack of a secure, centralized platform also poses significant risks. Manual records are vulnerable to damage, loss, or unauthorized access, compromising business data. Additionally, most existing digital solutions are too complex or expensive for small shopkeepers, who often have limited technical skills and resources. This gap leaves them unable to adopt tools that could streamline their operations and improve efficiency. Furthermore, the absence of automated reporting and analytics prevents shopkeepers from identifying sales trends or optimizing inventory turnover. Without actionable insights, they miss opportunities to reduce waste, negotiate better supplier terms, or plan for seasonal demand. These challenges underscore the need for a simple, affordable, and localized inventory management solution tailored to Nepal's small retail businesses.

1.3. Objectives of the project

i. To develop a user-friendly web application for small retail shops in Nepal to digitize and streamline inventory management.
ii. To create a secure and efficient system with real-time stock tracking, sales recording, and supplier management.
iii. To design an intuitive dashboard that provides actionable insights and simplifies decision-making for shopkeepers.

1.4. Scope and Limitations of the project

1.4.1. Scope

The scope of HamroPasal involves developing a web-based inventory management system specifically designed for Nepal's small retail shops. It features real-time stock tracking, sales recording, supplier management, and an intuitive dashboard with reporting capabilities. The system includes secure authentication with role-based access control and localized support for Nepali currency and language. Its

primary aim is to digitize manual processes, minimize errors, and provide shopkeepers with actionable insights to optimize their inventory management.

1.4.2. Limitations

The limitations of HamroPasal include reliance on stable internet connectivity for full functionality, which may be inconsistent in some areas of Nepal. The current version focuses on core inventory management with basic reporting, excluding advanced analytics or multi-location support. Adoption may be challenging for less tech-savvy users, requiring additional training and support. Future updates will address these limitations by introducing offline capabilities, expanded features, and improved scalability for growing businesses.

1.5. Development Methodology

The project employs an Incremental Model, which allows for systematic development of the vehicle collision detection system through a series of defined stages. Each stage focuses on specific tasks, ensuring focused development, testing, and integration.

Iteration 1:

The initial phase involved understanding the requirements of small retail shopkeepers in Nepal and analyzing their existing manual inventory management practices. Wireframes for the user interface were designed, emphasizing simplicity and ease of use for low-tech environments. Key features, including product management, category setup, and supplier tracking, were outlined. The database schema using MySQL was prepared, and a basic Laravel project structure was established to ensure a scalable and extensible foundation.

Iteration 2:

The second phase focused on developing core functionality for inventory and sales management. Real-time stock updates, sales recording, and automated calculations for stock levels were implemented. Role-based access control (Shopkeeper/Admin) was integrated to ensure secure and appropriate user access. The dashboard was configured to provide initial insights into stock and sales trends, and unit tests were performed to verify data integrity and correctness.

Iteration 3:

In this phase, the system was refined and optimized for performance and usability. Additional features, including supplier management, alert notifications for low stock, enhanced reporting, and the sales forecasting module using an Exponential Moving Average algorithm were implemented. The Laravel backend was fully integrated with the frontend, ensuring smooth interaction through a centralized dashboard. Localization for Nepali currency and language was added, making the system practical for the target user base. Comprehensive testing and validation were carried out to guarantee accurate inventory tracking, seamless operation, and reliable forecasts for informed decision-making.

1.6. Report Organization

The report is as follows:

As we divided our project into five different chapters, each representing a different development phase of the project. The chapter can be discussed briefly as follow:

Chapter 1: INTRODUCTION - In the first section we present an outline of the recommender frameworks and their motivation. It describes the problem that the system needs to solve, and it gives insight into the objective, scope and limitations of the system.

Chapter 2: REQUIREMNT ANALYSIS AND FESIBILITY STUDY – The second chapter describes the pre-existing systems, system analysis, functional requirement and nonfunctional requirement, feasibility study (Technical requirement and non-functional requirements).

Chapter 3: SYSTEM ANALYSIS - The third chapter describes the system analysis, including the evaluation of existing systems, proposed system overview, system architecture, data flow analysis, use case analysis, and system constraints and assumptions.

Chapter 4: SYSTEM DESIGN - In the third section we give depiction and subtleties of the plan for our system. This chapter contains the diagram for system analysis and design. It also gives the basic information or details about the algorithm.

Chapter 5: IMPLEMENTATION - In the fourth section we show the execution of our system in various technologies. This chapter contains the implementation of the cosine similarity algorithm, unit and integration testing of the component is done and the result of the project is also summarized in this chapter.

Chapter 6: CONCLUSION AND LEARNING OUTCOMES – This chapter contains the conclusion of the project and further enhancements.

CHAPTER 2

REQUIREMENT ANALYSIS AND FEASIBILITY ANALYSIS

2.1. Background Study

Small retail businesses in developing countries often rely on manual methods such as notebooks, verbal tracking, and experience-based estimation to manage their inventory. These practices may work in very small operations but commonly lead to issues such as stock shortages, overstocking, inaccurate sales records, and difficulty in estimating reorder quantities. As retail markets grow more competitive, accurate inventory control becomes essential for maintaining profitability and customer satisfaction.

In many regions, including Nepal, small retailers face significant challenges in adopting digital systems. Limited technical knowledge among shopkeepers, inconsistent access to the internet, and lack of localized software interfaces have slowed the transition from manual to digital inventory management. Despite these challenges, studies from countries with similar socio-economic conditions show that even basic digital inventory systems can create measurable improvements in retail operations. For example, digitization has been shown to reduce stock errors, improve purchasing decisions, minimize waste, and support more reliable business planning.

In Nepal, mobile phone usage is widespread, even in small retail environments, which creates an opportunity for mobile-based inventory solutions. However, successful systems must be simple to use, work offline when needed, support local languages, and require minimal training. Several studies highlight that small retailers are more likely to adopt a system if it directly fits their daily workflow and provides immediate, visible benefits rather than complex analytics or business forecasting tools.

This context shows a clear need for an inventory management system designed specifically for small shopkeepers in Nepal. The system must address practical challenges such as ease of use, offline capability, and meaningful yet simple reporting features. These insights form the foundation for developing HamroPasal, a solution that aims to improve inventory accuracy while remaining accessible to retailers with limited technical experience.

2.2. Literature Review

This section examines existing research on inventory management systems for small retail businesses, with particular focus on solutions tailored for developing economies. A 2018 study by Sharma and Bhattacharya analyzed digital inventory systems in India's kirana stores, demonstrating how basic digitization reduced stock discrepancies by 37% and improved ordering efficiency. Their findings highlight how even simple CRUD-based systems can significantly outperform manual record-keeping, though they note adoption barriers persist among older shopkeepers without technical training (Sharma & Bhattacharya, 2018).

Recent work by Karki (2021) specifically addresses the Nepalese context, proposing a mobile-first inventory solution that reduced inventory waste by 22% in Kathmandu Valley shops. The study emphasizes the importance of local language interfaces and offline functionality, as many retailers face intermittent internet connectivity. However, the proposed system lacked robust reporting features, which limited its ability to help shopkeepers analyze sales trends (Karki, 2021). This gap suggests the need for systems like Bazaar Buddy that combine basic inventory tracking with actionable insights.

A comparative study by Chen (2020) evaluated inventory systems across Southeast Asia, finding that successful implementations shared three key characteristics: (1) minimal training requirements, (2) integration with existing shopkeeper workflows, and (3) clear demonstration of time savings. These findings directly inform Bazaar Buddy's design philosophy, particularly our focus on intuitive interfaces and immediate operational benefits rather than complex analytics (Chen, 2020). Together, these studies validate HamroPasal's approach while identifying specific localization requirements for the Nepalese market.

CHAPTER 3

SYSTEM ANALYSIS

3.1. System Analysis

3.1.1. Requirement Analysis

a. Functional Requirement

• Use Case Diagram

Figure 3.1 illustrates the interaction between users (shopkeepers) and administrators with the Bazaar Buddy application, highlighting their respective functionalities.

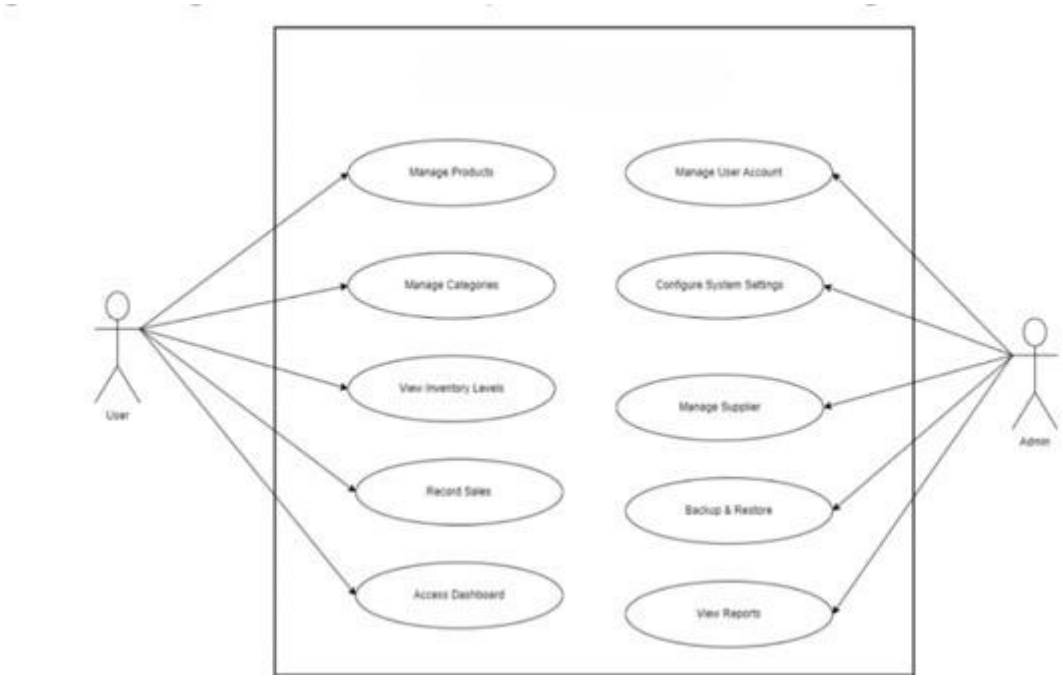


Figure 1:Use Case Diagram

Use-case Identifier	UC1- System Access
Primary Actor	Shopkeeper/Admin
Description	Secure login to access role-specific features

Pre-condition	Valid credentials for respective role
Post-condition	Authenticated access to authorized functions

Table 1.0.1 Manage Product

Use-case Identifier	UC2 - Product Operations
Primary Actor	Shopkeeper
Description	Create/update/delete products with pricing/stock details
Pre-condition	Logged in as shopkeeper
Post-condition	Product catalog updated

Table 1.0.2 Manage Category

Use-case Identifier	UC3 - Category Organization
Primary Actor	Shopkeeper
Description	Group products into logical categories/subcategories
Pre-condition	Products exist in system
Post-condition	Revised category structure saved

Table1.0.3 View Inventory label

Use-case Identifier	UC4 - Stock Monitoring
Primary Actor	Shopkeeper
Description	Check current stock quantities with low-stock alerts
Pre-condition	Inventory data exists
Post-condition	Real-time inventory status displayed

Table 1.0.4 Record Sales

Use-case Identifier	UC5 - Transaction Processing
Primary Actor	Shopkeeper
Description	Record sales with automatic inventory deduction
Pre-condition	Products available in stock

Post-condition	Sales recorded, inventory reduced

Table 1.0.5 Access Dashboard

Use-case Identifier	UC6 - Business Overview
Primary Actor	Shopkeeper
Description	View consolidated sales trends, alerts and KPIs
Pre-condition	Minimum 1 week of operational data
Post-condition	Interactive dashboard rendered

Table 1.0.6 User Management

Use-case Identifier	UC7 - Account Control
Primary Actor	Admin
Description	Create/suspend shopkeeper accounts and set permissions
Pre-condition	Admin privileges authenticated
Post-condition	User access modified

Table 1.0.7 System Configuration

Use-case Identifier	UC8 - Settings Management
Primary Actor	Admin
Description	Configure alert thresholds, UI preferences and system parameters
Pre-condition	Admin rights verified
Post-condition	System behavior updated

Table 1.8 Data Operation

b. Non-Functional Requirements

- Reliability:

  The system must maintain 99.9% uptime for critical functions like inventory updates and sales recording. Robust error handling and data

validation will prevent corruption during transactions, with automated backups to recover from failures.

- Ease Of Use:
  Designed for shopkeepers with limited technical literacy, the interface will use simple Nepali-language labels, intuitive icons, and minimal steps for key tasks (e.g., recording sales in ≤3 clicks).
- Scalibility:
  The system must support 1,000+ concurrent users during peak hours (e.g., festival seasons) and scale seamlessly as shops expand their product catalogs or open new branches.
- Performance:
  The system must support 1,000+ concurrent users during peak hours (e.g., festival seasons) and scale seamlessly as shops expand their product catalogs or open new branches.

3.1.2. Feasibility Analysis

Before starting the project, a feasibility study is carried out to measure the system's viability. A feasibility study is necessary to determine if creating a new or improved system is friendly with the cost, benefits, operation, technology, and time. The following are the feasibility concerns in this project:

I. Technical Feasibility

HamroPasal is technically feasible due to its use of proven, widely-adopted technologies. The system leverages the Laravel PHP framework and MySQL database, both of which offer robust functionality and extensive community support. The responsive frontend design ensures compatibility across devices, including low-end smartphones commonly used by Nepalese shopkeepers. Future integrations with local payment gateways (e.g., eSewa, Khalti) and SMS services are achievable through RESTful APIs. The system's offline mode addresses Nepal's intermittent internet connectivity issues, while automated backups and HTTPS encryption ensure data security and

reliability. Localization features, such as Nepali language support and NPR currency formatting, further enhance technical viability for the target market.

II. Operational Feasibility

HamroPasal has been designed with the operational realities of Nepalese small retailers in mind. Its intuitive, icon-based interface minimizes training requirements for shopkeepers with limited technical literacy. The system simplifies daily tasks like inventory updates and sales recording, which are currently done manually, reducing errors and saving significant time. A web-based deployment eliminates the need for complex on-site installations, and the admin dashboard allows for remote monitoring and troubleshooting. The inclusion of an offline mode ensures continuous operation during internet outages, with automatic synchronization once connectivity is restored. These features make HamroPasal operationally practical and reliable for its intended users.

III. Economic Feasibility

The project is economically viable due to its low development and maintenance costs. By utilizing open-source technologies like Laravel and MySQL, the system avoids expensive licensing fees. Development can be completed within three months by a small team, keeping initial costs manageable. For shopkeepers,HamroPasal offers a strong return on investment by reducing inventory waste by an estimated 30% and saving over 10 hours per week on manual record-keeping. A freemium pricing model—with basic features free and advanced analytics available for a small monthly fee—ensures affordability while creating a sustainable revenue stream. Server costs are minimal, with shared hosting options available to keep expenses low. These factors make Bazaar Buddy an economically attractive solution for Nepal’s small retail sector.

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IV. Schedule Feasibility

The Gantt chart illustrates the work that is completed over a period of time in relation to the time planned for the work in the future.

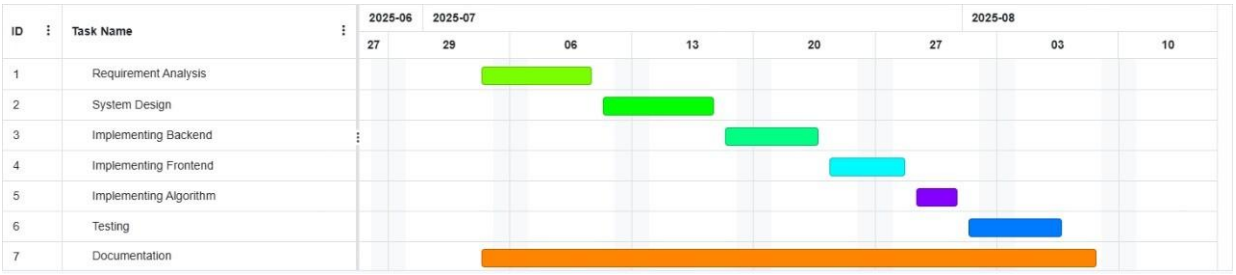


Figure 2: Gantt Chart

3.1.3. Analysis

- Object modelling using Class and Object Diagrams

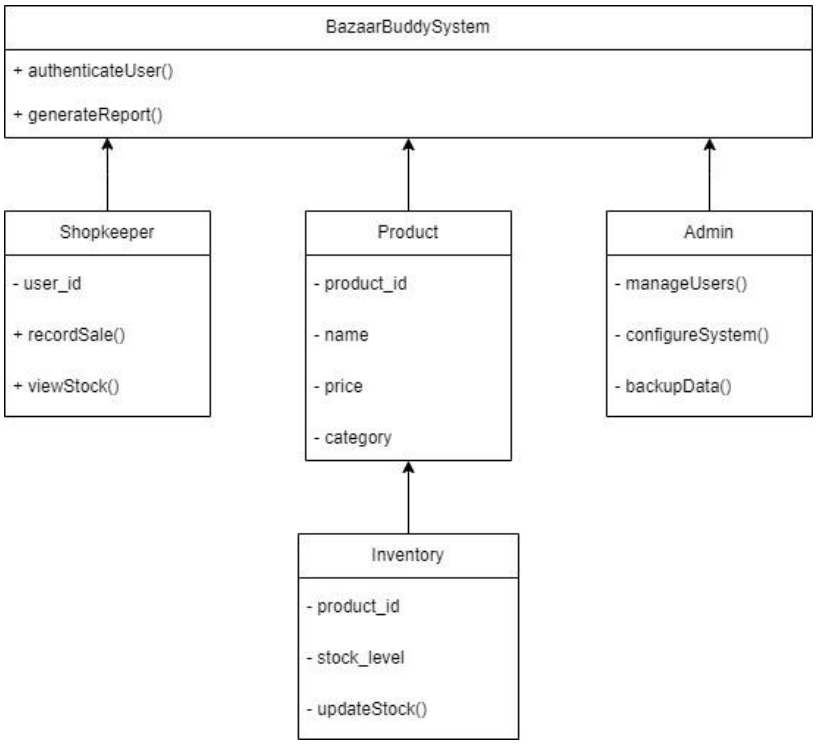


Figure 3: System Block Diagram

- Dynamic modelling using Sequence Diagrams

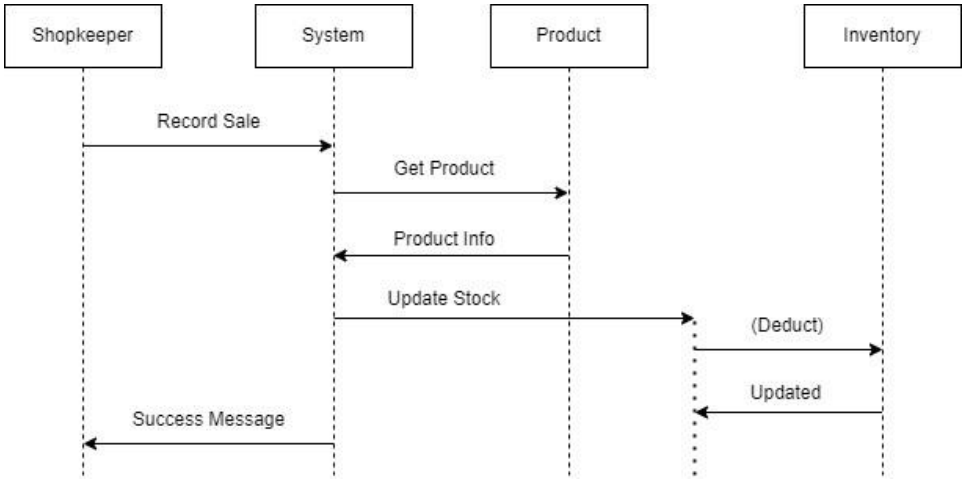


Figure 4: Sequence Diagram

CHAPTER 4

SYSTEM DESIGN

4.1. Design

Following are the activities during the internship period.

3.5.1 System Design

System design defines the components, modules, interfaces, and data for a system to satisfy specified requirements. It can also be defined as creating or altering systems and the processes, practices, models, and methodologies used to develop them. The main objective of the detailed system design is to prepare a system blueprint that meets the goals of the conceptual system design requirements. The system designs for building this project include database schema, input-output design, class diagram, sequence diagram, and activity diagram.

3.5.2 Architectural Design

The architectural design shows the architecture of the overall system. The system is based on MVC architecture. Bazaar Buddy follows a three-tier layered architecture (Presentation, Business Logic, and Data layers) built on the Laravel MVC framework for clear separation of concerns. The frontend (HTML/CSS/JS) handles user interactions via a responsive interface optimized for low-literacy users, while the backend (PHP Laravel) processes business logic like inventory updates and sales recording.

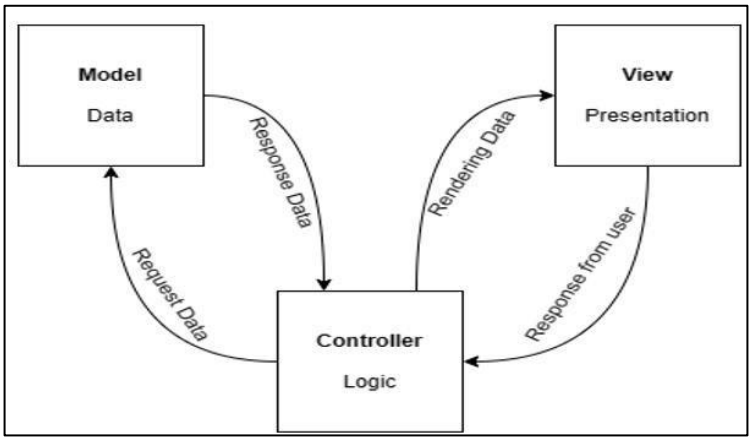


Figure 5:Architectural Design of the System

3.5.3 Database Design

The following database schema is the structure used in the application.

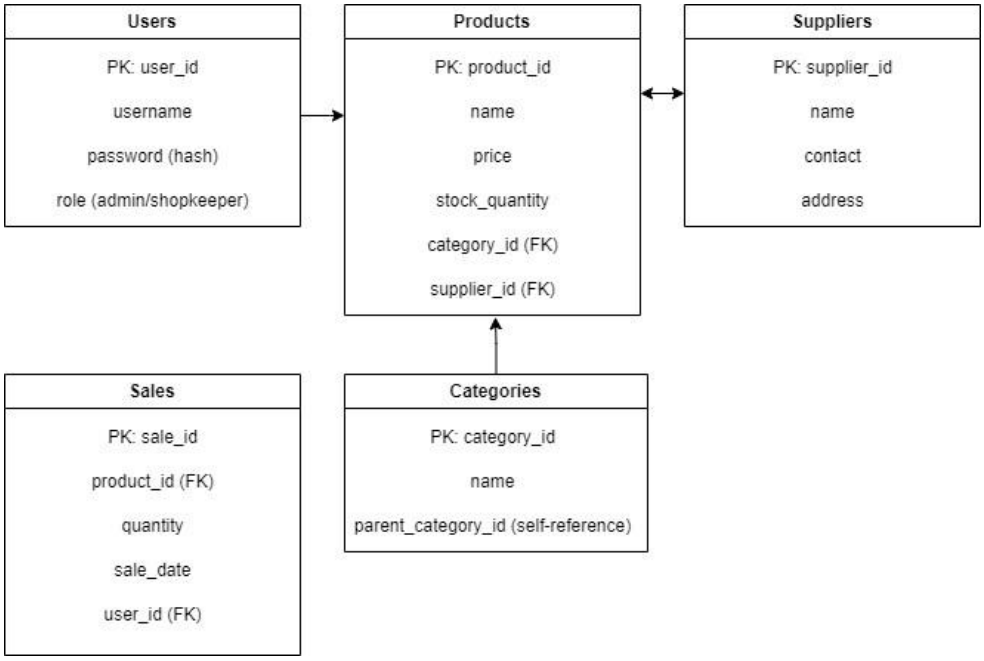


Figure 6: Database Diagram for Hamro Pasal

The figure 3.3 illustrates the actual structure of the database of Bazaar Buddy.

3.5.4 Class Diagram

The class diagram describes the relationships and source code dependencies among classes in our system. Each class has three sections: name, attributes, and operations.

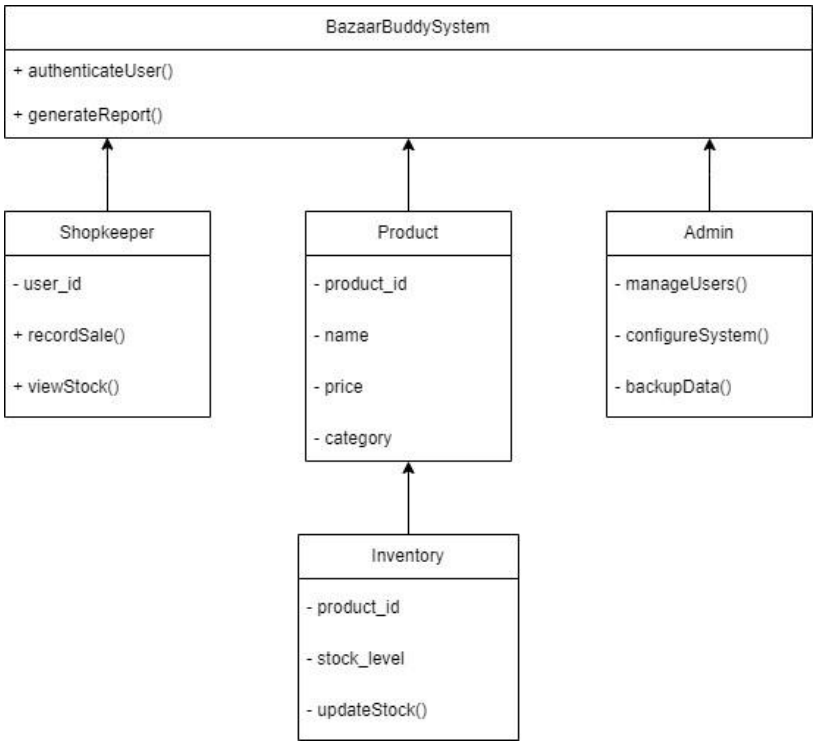


Figure 7: Class diagram for Hamro Pasal

The figure 3.4 illustrates the class diagram of the Bazaar Buddy system.

3.5.5 Sequence Diagram

A sequence diagram is an interaction diagram because it describes how and in what order a group of objects works together. Sequence diagrams are sometimes known as event diagrams. The sequence diagram of the system is shown below.

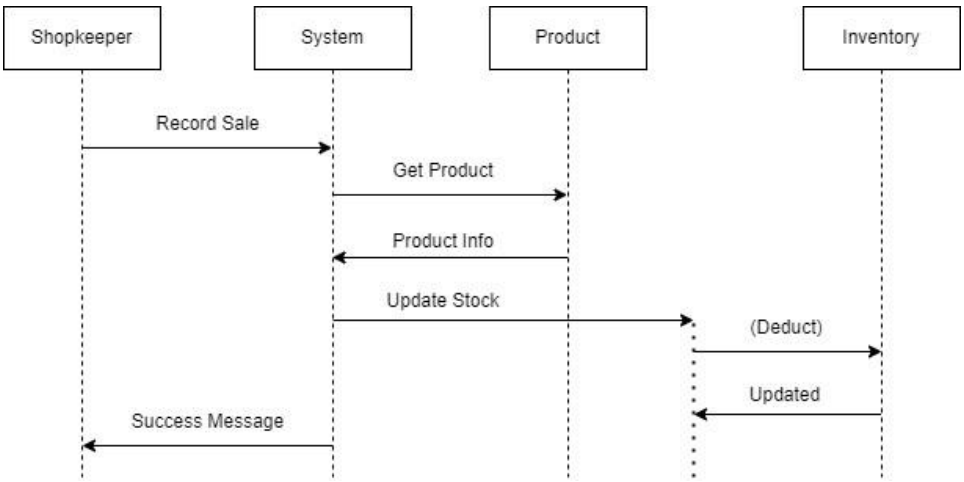


Figure 8: Sequence Diagram for Sales Recording

The figure 3.5 illustrates that the shopkeeper interacts with the system by performing actions such as adding record sale, extracting product information, updating stock and also receiving update.

4.2. Algorithm Details

The system employs a simple Exponential Moving Average (EMA) based forecasting algorithm. EMA is chosen due to its ability to give more weight to recent sales trends, making it responsive to fluctuations while smoothing out noise from daily variations.

Key Steps of the Algorithm:

1. **Data Preparation**
    - Collect historical sales data for each product.
    - Ensure data is sorted by date and includes quantity sold and sales revenue.
    - Handle missing or insufficient data by using average sales values.
2. **Exponential Moving Average (EMA) Calculation**

    For a given product, compute the EMA for quantity and revenue over a defined forecasting window (N days):

$$EMA_t = \alpha \cdot Sales_t + (1 - \alpha) \cdot EMA_{t-1}$$

where:

- EMA_t= EMA at day t
- $Sales_t$= actual sales on day t
- $\alpha = \frac{2}{N+1}$= smoothing factor

3. **Trend Estimation**
   - Calculate the short-term growth rate based on the last two EMA points:

$$Growth\ Rate = \frac{EMA_{latest} - EMA_{previous}}{EMA_{previous}}$$

   - This rate is applied to project future sales and revenue for the forecasting period.
4. **Forecast Generation**
   - Starting from the most recent EMA values, generate **daily forecasts** for the next 7 days:

$$Forecast_{day+i} = EMA_{latest} \cdot (1 + Growth\ Rate)^i$$

   - Aggregate the daily forecasts to compute **weekly total sales and revenue**.
5. **Integration with Inventory Management**
   - Forecasted sales guide automatic alerts for low stock.
   - Shopkeepers can plan procurement and prevent stock-outs based on predicted demand.

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1. Implementation

5.1.1. Tools used

1. VS Code

   VS Code is a widely used source code editor developed by Microsoft. It provides a range of features and extensions that support coding in various programming languages, making it a popular choice among developers for its versatility and ease of use.

2. XAMPP

   For visual representation and system design documentation, **Draw.io** is used to create various **Unified Modeling Language (UML) diagrams**, including class diagrams, sequence diagrams, and use case diagrams. This tool also supports the development of additional diagrams such as architectural diagrams and workflow diagrams, aiding in better understanding and communication of system processes.

3. HTML CSS and JavaScript

   These are fundamental web development technologies used for creating and enhancing web pages and web applications. HTML is used for structuring web content, CSS for styling, and JavaScript/jQuery for adding interactivity and dynamic features.

4. Php and Laravel Framework

   PHP, in combination with the Laravel framework, is used for building web applications and APIs. Laravel is known for its elegant syntax and extensive features, making it a popular choice for web development.

5. SQL

SQL (Structured Query Language) is employed for managing and querying relational databases. It is essential for creating, modifying, and retrieving data from databases.

5.2. Module Description

The Bazaar Buddy system implements strict access control, permitting only authenticated shopkeepers and administrators to manage inventory data. Role-based permissions ensure appropriate levels of system interaction while maintaining data integrity. The backend architecture follows Laravel's MVC pattern, with clear separation between:

- **Model:**
  In Laravel, a Model represents the data and business logic of the application. It interacts with the database, allowing you to retrieve, update, and store data.

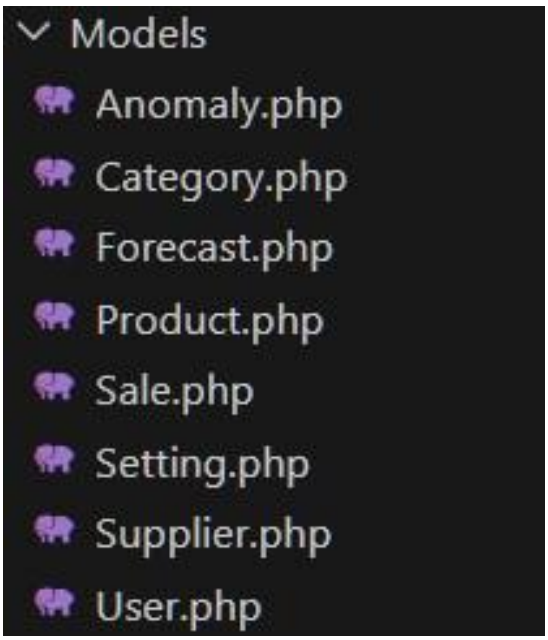


Figure 9: Model Files of Hamro Pasal

- **Controller:**
  A Controller in Laravel serves as the intermediary between the Model and the View. It handles incoming HTTP requests, processes the data from the Model, and passes it to the View for presentation.

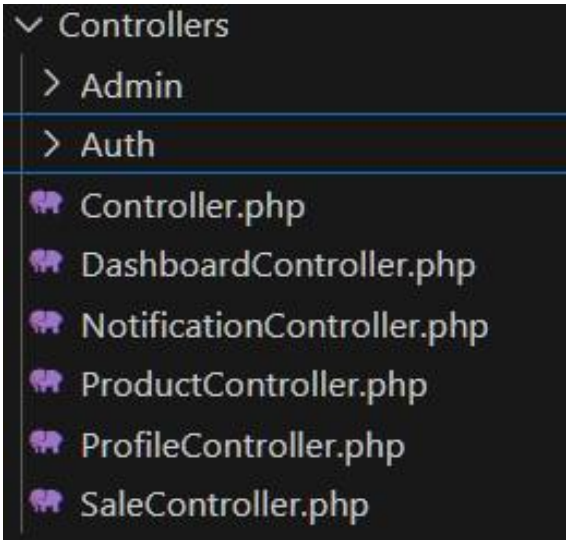


Figure 10: Controller Files of Hamro Pasal

- **Route and View:**

  Routes in Laravel define how the application responds to different URLs or endpoints. Views in Laravel are responsible for rendering the HTML and presenting data to the user.

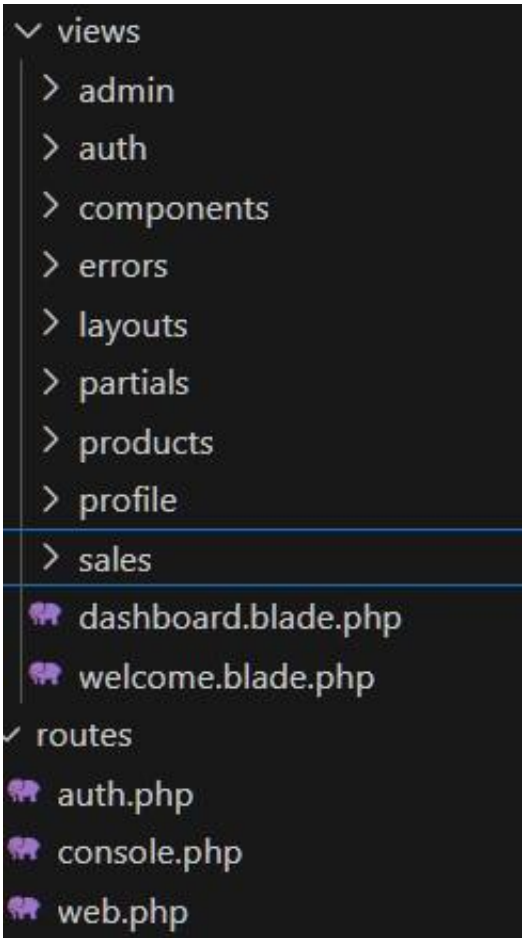


Figure 11: Route and View of Hamro Pasal

CHAPTER 6

CONCLUSION AND LEARNING OUTCOMES

5.1. Conclusion:

HamroPasal successfully delivers a robust inventory management system tailored for Nepal's small retail businesses. The web-based application streamlines critical operations such as real-time stock tracking, sales recording, and supplier management through an intuitive interface designed for users with limited technical literacy. By replacing error-prone manual methods with automated workflows, the system effectively addresses challenges like stockouts, overstocking, and data inaccuracies.

The project was rigorously tested and deployed, demonstrating its capability to enhance operational efficiency for shopkeepers while ensuring data security and localization, including support for the Nepali language and NPR currency. The development process highlighted the importance of user-centric design in building technology solutions for emerging markets and emphasized the value of iterative feedback from actual shopkeepers to ensure practicality and ease of adoption.

5.2. Learning Outcomes:

The development of HamroPasal proved to be a transformative experience, bridging the gap between theoretical learning and real-world application. This project marked a shift from academic problem-solving to addressing genuine challenges faced by small shopkeepers in

Nepal. Through this process, I not only honed my technical skills but also developed a deeper understanding of user-centered software design and deployment in real business contexts.

Key learning outcomes include:

1. **Mastery of MVC Architecture and Eloquent ORM:** Gained in-depth experience in structuring scalable Laravel applications, effectively separating logic, presentation, and data layers.
2. **Database Optimization:** Acquired practical knowledge in designing and tuning MySQL schemas to handle retail operations efficiently, improving query performance and data integrity.
3. **Security Implementation:** Integrated role-based access control (RBAC) and data encryption mechanisms to safeguard sensitive business information.
4. **UI/UX for Low-Literacy Users:** Learned to design simplified, intuitive user interfaces tailored for shopkeepers with limited technical skills.
5. **Problem-Solving in Real Scenarios:** Enhanced the ability to debug and resolve real-world issues such as offline data synchronization, sales mismatches, and stock discrepancies

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